

CO₂ - ACTIVITY - 2

1. CCTV image with salt & pepper noise

Technique: Median filter.

→ It removes salt & pepper noise effectively.

→ It also preserves important edges in the CCTV image.

2. Satellite image with Gaussian noise

Technique: Gaussian filter.

→ It reduces Gaussian noise caused by the sensors.

→ It smooths the image & improves its overall quality.

3. Medical x-ray with random noise

Technique: Bilateral filter.

→ It reduces random noise from the x-ray image.

→ It prevents fine edges & important medical structures.

4. Scanned document with black & white dots.

Technique: Median filter.

→ It removes scattered black & white noise dots.

→ It keeps the text & important details relatively clear.

5. Low-light image with heavy grain

Technique: Gaussian filter.

→ It reduces the grainy noise in the low-light image.

→ It produces a smoother & cleaner looking image.

6. Foggy road image with very low contrast

Technique: Contrast Stretching.

→ It increases the range of intensity values in the image.

→ This improves the visibility of the road & objects.

7. Dull & washed-out underwater image

Technique: Histogram Equalization.

→ It improves the contrast of the

underwater image.

→ It makes hidden details more visible & enhances the appearance.

8. Chest x-ray with poor visibility.

Technique: Histogram Equalization

→ It enhances the contrast of the chest x-ray.

→ This helps make internal structures more clearly visible.

9. Night-time traffic image with low intensity variation.

Technique: Contrast Stretching.

- It expands the intensity range of the dark image.
- This improves visibility of vehicles, roads & surroundings.

10. Grayscale image with narrow intensity range. Technique: Contrast stretching.

- It expands the narrow range of pixel intensities.

- This increases the overall contrast & improves image details.